

Gunjan Joshi, Ph.D

Machine-Learning Scientist, Helmholtz Zentrum Dresden Rossendorf, Germany



EDUCATION

- Oct 2021 – Sept 2024 **Doctor of Philosophy in Electrical Engineering and Information Systems**, The University of Tokyo, Japan
Degree awarded: 20 September 2024
Advisor: Prof. Akira Hirose and Associate Prof. Ryo Natsuaki (Co-Advisor)
Thesis title: *Inverse Mapping for Neural Network Explainability and its Applications in Earth Observation*
Focus Areas: Explainable Machine Learning, Remote Sensing, Multimodal Sensor Data Fusion, Cryosphere
Minors: *Global Future Faculty Development Program* (Center for Research and Development of Higher Education); *Global Leader Program for Social Design and Management* (Graduate School of Public Policy); *Science, Technology and Innovation Governance Education Program* (Graduate School of Public Policy), The University of Tokyo
- Oct 2019 – Sept 2021 **Master of Engineering in Electrical Engineering and Information Systems**, The University of Tokyo, Japan
Advisors: Prof. Akira Hirose and Associate Prof. Ryo Natsuaki
Thesis title: *Analysis of Significant Factors in Multi-Sensor Satellite Data Fusion for Earthquake Damage Assessment*
- Sept 2012 – July 2016 **Bachelor of Engineering in Electronics & Communication Engineering**, Visvesvaraya Technological University, India
Thesis title: *FPGA Implementation of Channel Emulator for Testing of Wireless Air Interface using VHDL*

PROFESSIONAL EXPERIENCE

- Oct 2024 – Present **Machine-Learning Scientist**, Helmholtz-Zentrum Dresden-Rossendorf, Germany
Supervisors: Dr. Peter Steinbach (*Department of Computational Science*) & Prof. Dr. Pedram Ghamisi (*Department of Exploration*)
Project: The 3D-ABC Foundation Model: Towards Global 3D Above and Below Ground Carbon Stocks [Helmholtz Foundation Model Initiative]
- Feb 2023 – March 2023 **Guest Scientist**, German Aerospace Center (DLR), Germany
Supervisor: Dr. Andreas J. Dietz (*Team Polar and Cold regions, German Remote Sensing Data Center*)
Project: Multi-sensor Multi-temporal Mapping of Swiss Alpine Glaciers using Explainable AI Techniques
- April 2018 – Sept 2019 **Research Scholar**, Electrical Engineering and Information Systems, The University of Tokyo, Japan
- Dec 2016 – March 2018 **RF Engineer**, TATA Advanced Systems, Bangalore, India
- July 2016 – Dec 2016 **Network Engineer**, Ericsson Global, Bangalore, India

RECOGNITIONS

Honors

- Oct 2025 **HZDR Prämie (Performance Bonus)** awarded in recognition of high commitment and excellent performance
- Nov 2023 **Rising Stars Women in Engineering** – Selected participant for The University of Tokyo

Awards

- July 2023 **Best Poster Award** from Department of Electrical Engineering and Information Sciences, The University of Tokyo.
- July 2022 **First Prize Three Minutes Thesis (3MT) Competition**, IEEE International Geoscience and Remote Sensing Symposium
- Nov 2021 **Japan Society of Photogrammetry and Remote Sensing (JSPRS) Award**, Asian Conference on Remote Sensing

Grants

- July 2023 **Remote Sensing Technology Center of Japan (RESTEC) Research Grant 2023**, PI: Gunjan Joshi, 1 million JPY
- IEEE-GRSS IDEA Professional Development Microgrants**
- Feb 2023 **The University of Tokyo Overseas Research Training (Musha Shugyo) Grant**

Scholarship

- April 2018 – Sept 2024 **Japanese Government (MEXT) Monbukagakusho** Scholarship

TEACHING AND TRAINING ACTIVITIES

- April 2025 **Course Instructor** for *Introduction to Machine Learning*, HIDA (Helmholtz Information and Data Science Academy)
- Oct 2025 **Course Instructor** for *Introduction to Deep Learning*, HIDA (Helmholtz Information and Data Science Academy)
- Nov 2025 **Trainer** for the *HPC-Gateway Fall School*, delivering sessions on Machine Learning
- Talk** on “*Use of Modern Large Language Models to Support Your Scientific Writing*” for the HZDR PhD Students’ Writing Sprint
- 2020–2024 **Teaching Assistant** for *Advanced Academic Writing and Presentation*, School of Engineering, The University of Tokyo
- Academic Writing Consultant** for *English Writing Center*, School of Engineering, The University of Tokyo

INVITED TALKS

- Jan 2026 **Invited Talk**, International Institute of Information Technology Bangalore (IIIT-B), Bengaluru, India. “*Opening the Black Box: Understanding Multi-Sensor Fusion through Explainable AI*”.
- Dec 2025 **Invited Speaker** for the IEEE GRSS Young Professionals Summit 2025, Colombo, Sri Lanka, to deliver talk on “*Advancing Earth Observation with Foundation Models*”.
- Oct 2025 **Invited Speaker** for the IEEE GRSS Sri Lanka Chapter, delivering the talk “*Making Sense of Sensors: Using Explainable AI for Multi-Sensor Data Fusion*”.

- Aug 2025 **Invited Speaker** in the IGARSS 2025 tutorial “*Machine Learning-Based Integration of Optical and SAR Data for Monitoring Earth’s Dynamic Processes*”, talk titled “*Explainable AI and Its Applications in Multi-Sensor Earth Observation*”.
- Dec 2024 **Invited Speaker** at the University of Melbourne, Australia for lecture on “*Explainable AI and Its Applications in Earth Observation*”.

ACADEMIC SERVICES

- Feb 2026 – Present **co-Editor**, Frontiers in Imaging
- March 2025 – Present **Editorial Board - Column Editor**, IEEE Geoscience and Remote Sensing Magazine
- July 2024 – Present **Webinar Strategist, Digital Content Team Lead**, IEEE Geoscience and Remote Sensing Society
- March 2023 – July 2024 **Social Media Ambassador**, IEEE Geoscience and Remote Sensing Society

REVIEWER

- 1 IEEE Open Journal of the Computer Society
- 2 IEEE Geoscience and Remote Sensing Magazine
- 3 European Journal of Remote Sensing
- 4 IEEE Transactions on Geoscience and Remote Sensing
- 5 Helmholtz AI Conference 2025
- 6 Elsevier Advances in Space Research
- 7 Asia-Pacific Conference on Synthetic Aperture Radar 2023
- 8 IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

PUBLICATIONS

Overall citation metrics

Total citations: 67, **h-index:** 5; **i10-index:** 4.

(Date accessed 02 Feb 2025)

Peer-reviewed Journal Publications

- 1 J. Steiner, W. Armstrong, W. Kochtitzky, R. McNabb, R. Aguayo, T. Bolch, F. Maussion, V. Agarwal, I. Barr, N. R. Baurley, M. Cloutier, K. DeWater, F. Donachie, Y. Drocourt, S. Garg, **G. Joshi***, B. Guzman, S. Kutuzov, T. Loriaux, C. Mathias, B. Menounos, E. Miles, A. Osika, K. Potter, A. Racoviteanu, B. Rick, M. Sterner, G. D. Tallentire, L. Tielidze, R. White, K. Wu, and W. Zheng, “Global mapping of lake-terminating glaciers,” in *Earth System Science Data* (preprint, in review), 2025, doi: 10.5194/essd-2025-315. *Journal Impact factor (2024): 11.6 ; 5-year Impact Factor : 13.9*
- 2 **G. Joshi***, C. A. Baumhoer, A. J. Dietz, R. Natsuaki and A. Hirose, “Multisensor Glacier Surface Classification Using Confidence-Aware Explainable Inverse-Mapping Neural Network,” in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024, doi: 10.1109/JSTARS.2024.3454789. *Journal Impact factor (2024): 5.3*
- 3 **G. Joshi***, R. Natsuaki and A. Hirose, “Application of Inverse Mapping for Automated Determination of Normalized Indices Useful for Land Surface Classification,” in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023, doi: 10.1109/JSTARS.2023.3308049. *Journal Impact factor (2024): 5.3, Citations: 10*

- 4 **G. Joshi***, R. Natsuaki and A. Hirose, "Neural-network fusion processing and inverse mapping to combine multi-sensor satellite data and analyze the prominent features," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023, doi: 10.1109/JSTARS.2023.3247788. [Journal Impact factor \(2024\): 5.3, Citations: 16](#)

Peer-reviewed International Conferences

- 5 G. Grosse, J. Hashemi, L. van Delden, T. Lübker, I. Nitze, J. Strauss, S. Kruse, P. Ghamisi, P. Steinbach, A. Rizaldy, **G. Joshi***, W. Yu, R. Gloaguen, G. Cavallaro, E. Zandi, R. Sedona, S. Hashim, S. Mandal, M. Herold, Q. Song, S. Besnard, M. Urbazaev, M. Sips, A. Huth, I. Hajnsek, and M. Pardini, "3D-ABC: A Foundation Model for Global Terrestrial 3D Above and Below Ground Carbon Stock Mapping," *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*, Brisbane, Australia, 2025, pp. 862–866, doi: 10.1109/IGARSS55030.2025.11242519.
- 6 J. Hashemi, S. Besnard, G. Cavallaro, P. Ghamisi, R. Gloaguen, I. Hajnsek, S. Hashim, M. Herold, U. Herzsuh, A. Huth, **G. Joshi***, S. Kruse, T. Lübker, S. Mandal, I. Nitze, M. Pardini, A. Rizaldy, F. Röseler, R. Sedona, M. Sips, Q. Song, P. Steinbach, J. Strauss, M. Urbazaev, L. van Delden, W. Yu, E. Zandi, and G. Grosse, "Towards Enhancing Permafrost Carbon Stock Estimates using 3D-ABC: a Multimodal Foundation Model for Geospatial Analytics," extended abstract, submitted to *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*, 2025.
- 7 G. Grosse, P. Ghamisi, G. Cavallaro, M. Herold, A. Huth, I. Hajnsek, and **the 3D-ABC Team***. "Towards a Foundation Model for Global Terrestrial 3D Above and Below Ground Carbon Stock Mapping (3D-ABC)," *European Geosciences Union (EGU) General Assembly*, Vienna, Austria, April 2025. [EGU25-19378]
- 8 **G. Joshi***, R. Natsuaki, and A. Hirose, "Complex-Valued Neural Network Inverse Mapping for Explainability in PolSAR/InSAR Applications," *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, Athens, Greece, 2024, pp. 2000–2003, doi: 10.1109/IGARSS53475.2024.10640643.
- 9 **G. Joshi***, C. A. Baumhoer, A. J. Dietz, R. Natsuaki, and A. Hirose. "Estimating Snow Line Altitude by Optical and SAR Data Fusion: Inverse-mapping Explainable Neural Network-Based Approach - Case Study of the Great Aletsch Glacier" *15th European Conference on Synthetic Aperture Radar (EUSAR 2024)*, Munich, Germany, 2024, pp 399-404. doi: <https://ieeexplore.ieee.org/document/10659607>.
- 10 **G. Joshi***, R. Natsuaki and A. Hirose, "Fusion of ALOS-2/PALSAR-2 and Sentinel-1 Polarimetric Imagery with Explainable Neural Network-based Confidence Level Estimation for Glacier Surface Classification," *2023 8th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*, Bali, Indonesia, 2023, pp. 1-6, doi: 10.1109/APSAR58496.2023.10388723. [Citations: 1](#)
- 11 **G. Joshi***, R. Natsuaki and A. Hirose, "Automated determination of Normalized Indices useful for Glacier Surface Classification," *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, Pasadena, CA, USA*, 2023, pp. 215-218, doi: 10.1109/IGARSS52108.2023.10281784. [Citations: 1](#)
- 12 **G. Joshi***, R. Natsuaki and A. Hirose, "Neural Network Model for Multi-Sensor Fusion and Inverse Mapping Dynamics for the Analysis of Significant Factors," *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium*, 2022, pp. 473-476, doi: 10.1109/IGARSS46834.2022.9884409. [Citations: 7](#)
- 13 **G. Joshi***, R. Natsuaki, and A. Hirose, "Multi-Sensor Satellite-Imaging Data Fusion for Earthquake Damage Assessment and the Significant Features," *Proc. - 2021 7th Asia-Pacific Conf. Synth. Aperture Radar, APSAR 2021*, 2021, doi: 10.1109/APSAR52370.2021.9688438. [Citations: 12](#)
- 14 **G. Joshi***, R. Natsuaki, and A. Hirose. "SAR and Optical Sensor Data Fusion for Earthquake Damage Assessment and Analysis of the Significant Features," *The 42nd Asian Conference on Remote Sensing (ACRS 2021)*, Can Tho, Vietnam. [ACRS Proceedings].
- 15 **G. Joshi***, P. R. Prasad and A. Singh, "FPGA Implementation of Channel Emulator for Testing of Wireless Air Interface Using VHDL," *2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, Bangalore, India, 2016, pp. 728–732, doi: 10.1109/RTEICT.2016.7807920. [Citations: 17](#)

Others

- 16 J. Hashemi, S. Besnard, G. Cavallaro, P. Ghamisi, R. Gloaguen, I. Hajnsek, S. Hashim, M. Herold, U. Herzschuh, A. Huth, **G. Joshi***, S. Kruse, T. Lübker, S. Mandal, I. Nitz, M. Pardini, A. Rizaldy, F. Röseler, R. Sedona, M. Sips, Q. Song, P. Steinbach, J. Strauss, M. Urbazaev, L. van Delden, W. Yu, E. Zandi, and G. Grosse. "Permafrost Soil Carbon Stock Estimation using a Multimodal Geospatial Foundation Model, 3D-ABC," *ESA–NASA International Workshop on AI Foundation Models for Earth Observation*, ESA-ESRIN, Frascati, Italy, 5–7 May 2025. [PDF]
- 17 A. Hirose and **G. Joshi***. "Advanced Earth Observations Using Explainable AI to Monitor Glaciers for Water Resource Evaluation," *IEEE Humanitarian Activities Workshop with AI Technologies, IEEE World Congress on Computational Intelligence (WCCI)*, 2024.
- 18 A. Hirose, R. Natsuaki, and **G. Joshi***. "Explainable Artificial Neural Networks in PolSAR Data Processing," *Joint PI Meeting of JAXA Earth Observation Missions FY2024*, 18–22 November 2024.
- 19 A. Hirose and **G. Joshi***. "Advanced Earth Observations Using Explainable AI to Monitor Glaciers for Water Resource Evaluation," *IEEE Humanitarian Activities Workshop with AI Technologies (IJCNN Workshop on WCCI2024)*, June 30 2024.
- 20 A. Hirose, R. Natsuaki, and **G. Joshi***. "Fusion of Synthetic-Aperture-Radar and Optical Data and Explainability in Neural Network Processing," *The Joint PI Meeting of JAXA Earth Observation Missions FY2023*, November 6 - 10, 2023.
- 21 **G. Joshi**, R. Natsuaki, and A. Hirose. "Satellite Data Fusion for Damage Assessment and Analysis of the Significant Features," *Institute of Electronics, Information and Communication Engineers Technical Committee on Space, Aeronautical and Navigational Electronics (IEICE-SANE)*, vol. 121, no. 154, SANE2021-29, pp. 37-42, August 2021 [IEICE Technical Report].